THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The perimeter of a rectangle is 128 cm and the length is 3 times the breadth. Its area is: 1

(a) 768 cm² **(b)** 1024 cm² **(c)** 784 cm² **(d)** 736 cm²

2. An isosceles trapezium has parallel sides 10 cm and 22 cm and non-parallel sides 10 cm. Its area is: 1

(a) 160 cm² **(b)** 138 cm² **(c)** 256 cm² **(d)** 128 cm²

3. The length of a rectangle is increased by 20% and its breadth decreased by 10%. If the original area is 1000 cm², the new area (in cm²) is: 1

(a) 1200 **(b)** 1100 **(c)** 1000 **(d)** 1080

4. The area of a triangle is 120 cm². If the altitude on the side of 20 cm is 12 cm, the altitude on the side of 16 cm (in cm) is: 1

(a) 7 **(b)** 17 **(c)** 15 **(d)** 12

5. A rhombus has perimeter 116 cm and one diagonal 40 cm. Its area is: 1

(a) 1160 cm² **(b)** 1680 cm² **(c)** 420 cm² **(d)** 840 cm²

6. A 52 m by 36 m rectangular lawn has a path 4 m wide inside along its boundary. The area of the lawn left is: 1

(a) 1232 m² **(b)** 1808 m² **(c)** 1536 m² **(d)** 1168 m²

7. A triangle has sides 33 cm, 56 cm and 65 cm. Its area is: 1

(a) 1072 cm² **(b)** 154 cm² **(c)** 1848 cm² **(d)** 924 cm²

8. The perimeter of a rectangle is 108 cm and the length is 5 times the breadth. Its area is: 1

(a) 387 cm² **(b)** 405 cm² **(c)** 729 cm² **(d)** 414 cm²

9. **Assertion (A):** The area of a parallelogram with base 12 cm and height 5 cm is 60 cm^2. 1

Reason (R): Area of a parallelogram = base × corresponding height.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The area of a triangle with base 10 cm and height 8 cm is 40 cm^2. 1

Reason (R): The area of a rectangle of length 10 cm and breadth 8 cm is 80 cm^2.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The perimeter of a rectangle is 100 cm and its length exceeds its breadth by 18 cm. Find its area. 2

12. A rectangle's length is increased by 30% and breadth decreased by 20%. By what percent does the area change? (original area 100 units) 2

13. The area of a triangle is 135 cm² and one side is 18 cm. Find the altitude on that side and the area of a triangle with the same base and double the altitude. 2

14. The perimeter of a rectangle is 120 cm and its length exceeds its breadth by 22 cm. Find its area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The perimeter of a rectangular plot is 160 m and its length is 3 times its breadth. Find its area and the cost of fencing at ₹ 15 per metre. 3

16. An isosceles trapezium has parallel sides 14 cm and 32 cm and each slant side 15 cm. Find its height and area. 3

17. A garden 40 m by 30 m has a rectangular pond 10 m by 8 m inside it and a path 2 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. 3

OR

The perimeter of a rectangular plot is 180 m and its length is 4 times its breadth. Find its area and the cost of fencing at ₹ 12 per metre. 3

18. An isosceles trapezium has parallel sides 20 cm and 44 cm and each slant side 20 cm. Find its height and area. 3

OR

A garden 50 m by 35 m has a rectangular pond 12 m by 9 m inside it and a path 3 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A trapezium-shaped field has parallel sides of 50 m and 30 m, and the distance between them is 20 m. The owner wants to level it. 5

(i) Find the area of the field.

(ii) Find the cost of levelling at ₹15 per m^2.

(iii) How many mats of area 4 m^2 are needed to cover the field?

(iv) If each parallel side is increased by 10 m with the same height, find the new area and the increase.

20. The area of a rhombus is 384 cm² and one diagonal is 24 cm. Find the other diagonal, the side, and the altitude. 5

OR

A trapezium ABCD has parallel sides AB = 18 cm and DC = 30 cm and height 8 cm. A parallelogram of base 18 cm and the same height is cut off from it, leaving a triangle. Find the area of the trapezium, the parallelogram and the remaining triangle. 5

END OF QUESTION PAPER — Total: 40 Marks